

PENG HAO

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Experience

Slalom Build

Machine Learning Architect

Sydney, AU

May. 2022 – Present

- Designed and deployed a serverless GenAI solution with AWS Glue, Bedrock and OpenSearch for a legal client, significantly enhancing productivity in analyzing customer survey data.
- Engineered and deployed an full-stack GenAI app with streamlit as frontend and AWS Fargate as severless backend for an internal initiative, which is now widely adopted as demo tool for pre-sales activities.
- Architected and developed a serverless GenAI Proof of Concept (PoC) featuring a React-based frontend and a backend utilizing AWS services such as API Gateway, Lambda, Bedrock, and OpenSearch. The solution enables automated drafting of email bodies using knowledge from an internal database and is fully provisioned via AWS CloudFormation for seamless deployment.
- Spearheaded the architecture and development of LLM-powered Proof of Concepts (PoCs) for diverse use cases, including automated testing for database modernization and agent evaluation, designed to demonstrate their potential and efficiency.
- Designed and developed a scalable and reproducible model training pipeline on AzureML for a transportation client, addressing complex object detection challenges.
- Delivered a comprehensive end-to-end MLOps solution for a manufacturing client using AWS SageMaker and GitHub Actions, leveraging Terraform for automated infrastructure provisioning.
- Refactored a segmentation model code using Ray and orchestrated pipeline through composer on GCP for a retail client, reducing the operation time by threefold.

Endeavour Group

Senior Data Scientist

Sydney, AU

Nov. 2020 – Feb 2022

- Operated and maintained a recommendation model for a major Australian liquor brand, generating personalized offer emails to millions of customers.
- Monitored and analyzed model and email campaign performance, providing recommendations for model optimization.
- Integrated experiment tracking into the existing ML pipeline, enabling rapid iteration and efficient model development for data scientists.

Elula

Data Scientist

Sydney, AU

Jun. 2018 – Nov 2020

- Developed a scalable data quality check tool using PySpark to handle billion-record datasets and detect data drift for continous model training.
- Designed and customized predictive models for predicting customer churn on home loan application.
- Engineered an automated data pipelines (ETL) on AWS for enhancing data analysis workflows.

Professional Certifications

- Databricks Certified Generative AI Engineer Associate
- Databricks Certified Machine Learning Professional
- AWS Certified Mahicne Learning - Specialty
- Google Cloud Professional Machine Learning Engineer

- Google Cloud Professional Data Engineer
- Google Cloud Professional Cloud Developer

Skills

Machine Learning, MLOps, GenAI, Python, AWS, GCP, Databricks, Azure, DevOps

Education

University of Technology Sydney, Computer Science
Doctor of Philosophy

Sydney, AU
Dec. 2017

- Thesis: Cross-domain Recommender Systems through Tag-based Models